DEEPAK PATKAR

PhD, Theoretical and Computational Chemistry

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SUMMARY

Theoretical and computational chemist with a PhD from Dr. Harisingh Gour University, India, and expertise in noncovalent interactions, method development, and molecular clusters. Currently a Postdoctoral Fellow at the Fujian Institute of Research on the Structure of Matter, Chinese Academy of Sciences (FJIRSM), working on metalloproteins. Recipient of the NAWA Ulam Postdoctoral Fellowship (Poland, 2027), CAS Postdoctoral Fellowship (China, 2025), University PhD Fellowship (India, 2019), and DST-INSPIRE Scholarship (India, 2014). Author of 11 research articles (5 first-author, 3 sole-author) in ACS, RSC, and Wiley journals; developer of iHBEQuant, a program automating MTA to quantify H-bond energies and cooperativity.

EXPERIENCE

Post-doctoral Fellow **2025-Present**
Fujian Institute of Research on the Structure of Matter, Chinese Academy of Sciences, Fuzhou China
Research on copper-peptide-mediated liquid-liquid phase separation using molecular dynamics simulations. Applying supervised machine learning to materials screening.

Senior Associate at Innodata India Private Limited **2024-2025**
Role: Subject Matter Expert
Contributed to image-based training data curation for large language models, including data annotation, content evaluation, and quality assurance.

Teaching Assistant at Dr. Harisingh Gour Vishwavidyalaya, Sagar, India **2021-2023**
Delivered undergraduate chemistry laboratory instruction and developed computational chemistry assignments. Guided students in scientific computing, data analysis, and interpretation of results.

ACADEMICS

Doctor of Philosophy, Chemistry **2019-2024**
Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India
Thesis Supervisor: Dr. Milind Madhusudan Deshmukh [link]
Thesis: *Quantum Chemical Studies on the Mixed Molecular Clusters: Quantitative Estimates of Hydrogen Bond Strength and Cooperativity*. [link]
Investigated individual hydrogen-bond energetics and cooperativity in mixed molecular clusters using quantum chemical methods. Applied the molecular tailoring approach and electronic structure analysis to quantify interaction energies and characterise noncovalent interactions.

Master of Science, Chemistry **2017-2019**
Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India
Project Title: Application of HPLC and Gas Chromatography for Chemical Analysis.
Project Supervisor: Dr. Abhilasha Durgavanshi [link]
Completed a research project on chromatographic methods for chemical analysis, focusing on high-performance liquid chromatography and gas chromatography.

Bachelor of Science, Triple major in Physics, Chemistry, Criminology and Forensic Science **2014-2017**
Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India
Built foundation in core principles and analytical techniques across physics, chemistry, criminology, and forensic science.

ACHIEVEMENTS

NAWA Ulam Fellowship (2027) | CAS Postdoctoral Fellowship (2025) | Best Oral Presentation (2023) | University PhD Fellowship (2019) | INSPIRE Scholarship (2014)

RESEARCH INTEREST

Areas of interest include theoretical and computational chemistry, quantum chemical methods, density functional theory, noncovalent interactions, transition-metal catalysis, molecular and biomolecular simulations, atomistic machine learning, cheminformatics and CADD, inverse molecular design and software development.

PROGRAMMES

iHBEQuant | CuBind | MetSiteScanner [GitHub]

SELECTED PUBLICATIONS

1. **Patkar, D.** From Weak to Functional: Network Cooperativity Effects in Cyclic Neutral $(HF)_m(NH_3)_n$ Hydrogen Bonded Clusters, *J. Phys. Chem. A* **2026**, 28, 25990–26000. [DOI]
2. Ismail, T. M.; **Patkar, D.**; Sajith, P. K.; Deshmukh, M. M. Hydrogen bond strengths in microhydrated clusters of HNO and HONO: Energetic insights via molecular tailoring approach *New J. Chem.* **2024**, 48, 11591–11605. [DOI]
3. Ismail, T. M.; **Patkar, D.**; Sajith, P. K.; Deshmukh, M. M. Interplay of Hydrogen, Pnicogen and Chalcogen Bonding in $X(H_2O)_{n=1-5}$ (X = NO, NO^+, and NO^-) Complexes: Energetics Insights via Molecular Tailoring Approach *J. Phys. Chem. A* **2023**, 127, 10360–10374. [DOI]
4. **Patkar, D.**; Chopra, D.; Deshmukh, M. M. Characterization of non-covalent contacts in mono- and di-halo substituted acetaldehydes: probing the substitution effects of electron donating and withdrawing groups *Phys. Chem. Chem. Phys.* **2023**, 25, 2946. [DOI]

5. **Patkar, D.**; Ahirwar, M. B.; Deshmukh, M. M. A Tug of War between the Self- and Cross-associating Hydrogen Bonds in Neutral Ammonia-Water Clusters: Energetic Insights by Molecular Tailoring Approach. *ChemPhysChem* **2022**, 23, e202200476. [DOI]
6. **Patkar, D.**; Ahirwar, M. B.; Deshmukh, M. M. Energetic Ordering of Hydrogen Bond Strengths in Methanol-Water Clusters: Insights via Molecular Tailoring Approach *ChemPhysChem* **2022**, 23, e202200143. [DOI]
7. **Patkar, D.**; Ahirwar, M. B.; Shrivastava, S. P.; Deshmukh, M. M. Assessment of Hydrogen Bond Strengths and Cooperativity in Self- and Cross-Associating Cyclic $(HF)_m(H_2O)_n$ (m + n = 2 to 8) Clusters. *New J. Chem.* **2022**, 46, 2368–2379. [DOI]
8. **Patkar, D.**; Ahirwar, M. B.; Gadre, S. R.; Deshmukh, M. M. Unusually Large Hydrogen-Bond Cooperativity in Hydrogen Fluoride Clusters, $(HF)_n$, n = 3 to 8, Revealed by the Molecular Tailoring Approach. *J. Phys. Chem. A* **2021**, 125, 8836–8845. [DOI]
9. Ahirwar, M. B.; **Patkar, D.**; Yadav, I.; Deshmukh, M. M. Appraisal of Individual Hydrogen Bond Strengths and Cooperativity in Ammonia Clusters via Molecular Tailoring Approach. *Phys. Chem. Chem. Phys.* **2021**, 23, 17224–17231. [DOI]
10. **Patkar, D.** Uncovering the Individual Hydrogen Bond Strengths and Cooperativity in Pure $(H_2S)_n$ and Mixed $(H_2O)_m(H_2S)_n$ (m + n = 2 to 8) Clusters. *ChemRxiv* **2025**. [DOI] **[Pre-print]**
11. **Patkar, D.** $HF\cdots H_2S$ Hydrogen-Bonded Clusters as a Model for Probing Subtle Cooperative Effects. *ChemRxiv* **2026**. [DOI] **[Pre-print]**

SELECTED TALKS

1. National Seminar on *Recent Advances in Chemical Sciences*, Department of Chemistry, Dr. Harisingh Gour Vishwavidyalaya Sagar, March, **2023**.
   Title: Competition between Self- and Cross-associating Hydrogen Bonds in Neutral Ammonia-Water Clusters: Energetic Insights by Molecular Tailoring Approach.
2. DAE Symposium on *Current Trends in Theoretical Chemistry*, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September, **2022**.
   Title: Energetic Ordering of Hydrogen Bond Strengths in Methanol-Water Clusters: Insights via Molecular Tailoring Approach.
3. DAE Symposium on *Current Trends in Theoretical Chemistry*, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September, **2021**.
   Title: Evaluation of individual H-Bond Strengths and Cooperativity in $(HF)_m(H_2O)_n$, (m + n) = 2 to 8 Clusters.
4. Theoretical Chemistry Symposium, IISER-Kolkata, IACS Kolkata, December, **2021**.
   Title: Estimation of Individual Hydrogen Bond Strengths and Cooperativity in $(HF)_m(H_2O)_n$, (m + n) = 2 to 8 Clusters via Molecular Tailoring Approach.
5. International Seminar, Dr. Harisingh Gour Vishwavidyalaya Sagar, January, **2021**.
   Title: Theoretical Study of Hydrosilylation of Olefin Catalyzed by Cationic Calcium Hydride Catalyst.

SELECTED CONFERENCES

1. National Seminar on *Recent Advances in Chemical Sciences*, Department of Chemistry, Dr. Harisingh Gour Vishwavidyalaya Sagar, March **2023**.
2. DAE Symposium on *Current Trends in Theoretical Chemistry*, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September **2022**.
3. Hand on Workshop *Molecular Tailoring Approach*, Department of Scientific Computing, Modelling and Simulation, Savitribai Phule Pune University, May **2022**.
4. Theoretical Chemistry Symposium, IISER-Kolkata, IACS Kolkata, Kalyani University, December, **2021**.
5. DAE Symposium on *Current Trends in Theoretical Chemistry'*, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September **2021**.

TECHNICAL SKILLS

My technical skills include quantum chemistry, DFT, molecular modelling, molecular dynamics simulations, scientific programming (Python, Bash, FORTRAN), Linux, and high-performance computing (HPC). I also have knowledge of atomistic machine learning, inverse molecular design, cheminformatics, bioinformatics, data analytics, and basic quantum computing. Software and tools include Q-Chem, PySCF, CP2K, LAMMPS, RASPA, PyMatGen, RDKit, Discovery Studio, PyMOL, ChimeraX, VESTA, OVITO, ChemCraft, Spartan, GaussView, ChemDraw, LaTeX, TensorFlow, and PyTorch (basic knowledge).

LANGUAGE

English: Advanced | **Hindi:** Native | **Sanskrit:** Intermediate | **Mandarin Chinese:** Beginner

REFERENCES

1. **Dr. Milind Deshmukh** | Ph.D. Supervisor
   Associate Professor, Department of Chemistry, University of Delhi, Delhi
   Mobile: +91 8989180813 | milind.deshmukh@gmail.com | [link]
2. **Prof. Deepak Chopra**
   Crystallography and Crystal Chemistry Laboratory, Department of Chemistry, IISER Bhopal
   Office: +91-755-669-1311; Fax: +91 755 6692392 | dchopra@iiserb.ac.in | [link]
3. **Dr. Pookkottu K. Sajith**
   Assistant Professor, Department of Chemistry, Farook College, Kozhikode, Kerala
   Mobile: +91 9349280803 | pksajith@farookcollege.ac.in | [link]